

Serosus layer

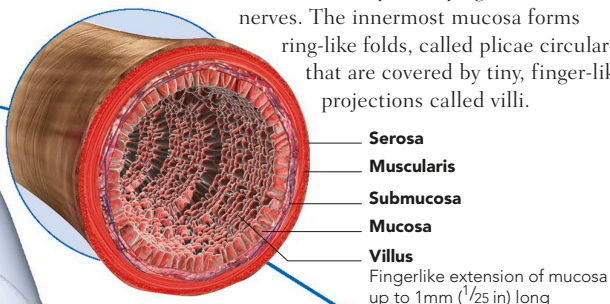
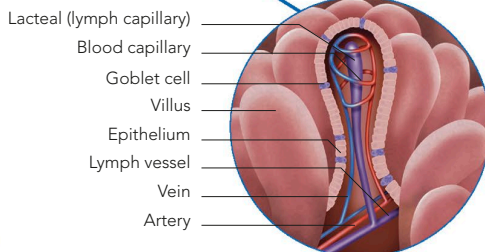
Clear membrane that covers the stomach externally

ROLE OF THE SMALL INTESTINE

The duodenum, jejunum, and ileum make up the small intestine. The duodenum receives not only processed food, known as chyme, from the stomach but also digestive secretions from the liver (including bile) and the pancreas. The jejunum and ileum are both long and coiled, but the jejunum is thicker, redder, and slightly shorter. In the small intestine, the chyme is broken down further by pancreatic juices, bile, and the intestine's own secretions, so that nutrients can be absorbed into the blood and lymph circulations.

LAYERS OF THE SMALL INTESTINE WALL

The small intestine wall has four layers. The outermost serosa is a protective coat. Next is the muscularis, which has outer longitudinal muscle fibres and inner circular smooth fibres. Inside this is the submucosa, a loose layer carrying vessels and nerves. The innermost mucosa forms ring-like folds, called plicae circulares, that are covered by tiny, finger-like projections called villi.

**SECTION OF SMALL INTESTINE****Jejunum**

About 2–2.5m (6½–8ft) long

Ileum

Third and longest section of the small intestine, up to 3.5m (11½ ft) in length

STOMACH AND SMALL INTESTINE

The stomach is situated in the upper left abdomen, protected by the lower ribs. The extensive small intestine lies looped and folded beneath it, and occupies most of the lower abdomen.

INTESTINAL VILLI

The epithelium of each villus lets digested nutrients pass into the interior, or lumen. Here, some pass into the lymph via a lacteal, and others enter the blood via a capillary and are carried to the liver. Epithelial cells have tiny microvilli, which increase the surface area of the small intestinal lining. Goblet cells scattered throughout the epithelium secrete mucus that helps the passage of food.